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Diameter-enlarged hole portion forming device

Abstract

A device for forming a diameter-enlarged hole portion by cutting into a hole that is formed in a workpiece, the device including: a bow jaw including a shaft and a guide, the guide including a distal end surface and an sloped surface, the distal end surface being provided on an opposite side of the guide from the shaft, the sloped surface being sloped from the shaft side of the guide toward the distal end surface side of the guide in a direction away from a center axis of the shaft; and a cutter blade including a cutting portion whose distal end is provided with a cutting edge, the cutting portion being slidable along the sloped surface. The bow jaw further includes a suction passage that includes a flow passage, the flow passage including a suction port that is open at the distal end surface.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to diameter-enlarged hole portion forming devices.

BACKGROUND ART

(2) Patent Literature 1 discloses a known conventional diameter-enlarged hole portion forming device for forming a diameter-enlarged hole portion by cutting into an anchor bolt implanting hole. At the time of forming a diameter-enlarged hole portion by cutting into a hole wall by using the diameter-enlarged hole portion forming device of Patent Literature 1, a hole wall cutting blade of the device moves along a guide groove. As the hole wall cutting blade moves forward along the guide groove, the hole wall cutting blade protrudes toward the outer side of a guide head, thereby cutting into the hole wall to form a diameter-enlarged hole portion.

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Utility Model Registration No. 2564100

SUMMARY OF INVENTION

Technical Problem

(4) In the case of the diameter-enlarged hole portion forming device of Patent Literature 1, powder dust is generated during the cutting by the hole wall cutting blade. The powder dust causes deterioration of the work environment, and if the powder dust enters between components of the diameter-enlarged hole portion forming device, it may cause malfunction of the device, wear of the components of the device, and deterioration in the performance of the diameter-enlarged hole portion forming device.

(5) The present invention has been made in order to solve the above-described problems. An object of the present invention is to provide a diameter-enlarged hole portion forming device capable of efficiently removing the powder dust, for example, as a measure for preventing the scattering of the powder dust and reducing the exposure to the powder dust.

Solution to Problem

(6) A diameter-enlarged hole portion forming device according to one aspect of the present invention is a diameter-enlarged hole portion forming device for forming a diameter-enlarged hole portion by cutting into a hole that is formed in a workpiece. The diameter-enlarged hole portion forming device includes: a bow jaw including a shaft and a guide, the guide including a distal end surface and an sloped surface, the distal end surface being provided on an opposite side of the guide from the shaft, the sloped surface being sloped from the shaft side of the guide toward the distal end surface side of the guide in a direction away from a center axis of the shaft; and a cutter blade including a cutting portion whose distal end is provided with a cutting edge, the cutting portion being slidable along the sloped surface. The bow jaw further includes a suction passage that includes a flow passage, the flow passage including a suction port that is open at the distal end surface.

(7) According to the above configuration, during the cutting to form the diameter-enlarged hole portion, since the distal end surface is inserted in the deeper side of the hole, the powder dust that is generated due to the cutting is efficiently sucked in through the suction port of the distal end surface, and thereby the powder dust can be removed from the hole.

(8) In the diameter-enlarged hole portion forming device, the guide may include: a pair of the sloped surfaces provided such that the center axis is interposed between the sloped surfaces; and a recess that is open at the distal end surface and an outer peripheral surface between the pair of the sloped surfaces.

(9) According to the above configuration, the powder dust enters the recess through its opening at the outer peripheral surface. This makes it possible to prevent a situation in which: the powder dust gets caught between components of the diameter-enlarged hole portion forming device; and thereby malfunctioning of the diameter-enlarged hole portion forming device is caused and/or wear of the

components is caused. Moreover, the powder dust that has entered the recess is led by the recess to its opening at the distal end surface, and then sucked in through the suction port of the distal end surface. In this manner, the powder dust can be efficiently removed by suction.

(10) In the diameter-enlarged hole portion forming device, the suction passage may include a first branch passage that is branched off from the flow passage and that communicates with the recess.

(11) According to this configuration, the powder dust that has entered the recess is sucked in through the suction port of the first branch passage, and thereby the powder dust can be removed efficiently.

(12) In the diameter-enlarged hole portion forming device, the guide may include a first groove that is connected to the suction port and to an opening of the recess, the opening being formed at the distal end surface.

(13) According to the above configuration, the powder dust that has entered the recess is led from the opening of the recess at the distal end surface to the suction port by the first groove, and sucked in through the suction port. In this manner, the powder dust can be efficiently removed by suction.

(14) In the diameter-enlarged hole portion forming device, the suction passage may include a second branch passage that is branched off from the flow passage and that is open at the sloped surface.

(15) According to the above configuration, the powder dust on the sloped surface can be sucked in through the suction port of the second branch passage, and thereby the powder dust can be removed efficiently.

(16) In the diameter-enlarged hole portion forming device, the guide may include a second groove that is open at the sloped surface and that is connected to the suction port at the distal end surface.

(17) According to the above configuration, the powder dust on the sloped surface is led to the suction port by the second groove, and thereby the powder dust can be removed efficiently by suction.

Advantageous Effects of Invention

(18) The present invention is configured as described above, and has an advantageous effect of being able to provide a diameter-enlarged hole portion forming device that is capable of efficiently removing powder dust.

(19) The above object, other objects, features, and advantages of the present invention will be made clear by the following detailed description of a preferred embodiment with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A is a view of a diameter-enlarged hole portion forming device according to an embodiment of the present invention as seen from a recess side; and FIG. 1B is a fragmentary sectional view showing a part of the diameter-enlarged hole portion forming device of FIG. 1A.

(2) FIG. 2A is a view of the diameter-enlarged hole portion forming device of FIG. 1A as seen from a sloped surface side; and FIG. 2B is a fragmentary sectional view showing a part of the diameter-enlarged hole portion forming device of FIG. 2A.

(3) FIG. 3A is a side view of a shank; and FIG. 3B is a sectional view of the shank.

(4) FIG. 4A is a side view of a cutter blade; and FIG. 4B is a view of cutting portions of the cutter blade as seen from the distal end side.

(5) FIG. 5A is a view of a bow jaw as seen from the recess side; FIG. 5B is a view of the bow jaw of FIG. 5A as seen from the sloped surface side; and FIG. 5C is a view of the bow jaw of FIG. 5A as seen from the distal end side.

(6) FIG. 6A is a sectional view of a stopper sleeve taken along a direction orthogonal to an axial

direction; and FIG. 6B is a sectional view of a pin.

(7) FIG. 7A is a view, as seen from the recess side, of a bow jaw of the diameter-enlarged hole portion forming device according to Variation 1 of the embodiment of the present invention;

(8) FIG. 7B is a view of the bow jaw of FIG. 7A as seen from the sloped surface side; and FIG. 7C is a view of the bow jaw of FIG. 7A as seen from the distal end side.

(9) FIG. 8A is a view, as seen from the recess side, of a bow jaw of the diameter-enlarged hole portion forming device according to Variation 2 of the embodiment of the present invention;

(10) FIG. 8B is a view of the bow jaw of FIG. 8A as seen from the sloped surface side; and FIG. 8C is a view of the bow jaw of FIG. 8A as seen from the distal end side.

(11) FIG. 9A is a view, as seen from the recess side, of a bow jaw of the diameter-enlarged hole portion forming device according to Variation 3 of the embodiment of the present invention;

(12) FIG. 9B is a view of the bow jaw of FIG. 9A as seen from the sloped surface side; and FIG. 9C is a view of the bow jaw of FIG. 9A as seen from the distal end side.

DESCRIPTION OF EMBODIMENTS

Embodiments

(13) <Configuration of Diameter-Enlarged Hole Portion Forming Device>

(14) As shown in FIG. 1A to FIG. 2B, a diameter-enlarged hole portion forming device **10** according to an embodiment of the present invention is a cutting device for forming a diameter-enlarged hole portion E in a hole (a pre-formed hole H) that is pre-formed in a workpiece B.

(15) The diameter-enlarged hole portion forming device **10** includes a shank **20**, a cutter blade **30**, and a bow jaw **40**. The cutter blade **30** is coupled to the shank **20**. The bow jaw **40** is inserted in a hole formed in the shank **20** and a hole formed in the cutter blade **30**. The shank **20** and the bow jaw **40** are fixed to a stopper sleeve **50**, and a suction adapter **60** is mounted to the shank **20**.

(16) It should be noted that the diameter-enlarged hole portion forming device **10** has a center axis G. The center axis G also serves as the center axis of each of the shank **20**, the cutter blade **30**, and the bow jaw **40**, and is equivalent to a rotational axis about which each of the shank **20**, the cutter blade **30**, and the bow jaw **40** rotates. The diameter-enlarged hole portion forming device **10** has a distal end side and a proximal end side. In the description herein, the distal end side of the diameter-enlarged hole portion forming device **10** is defined as the side that is closer to the cutter blade **30** than the shank **20** is and that encompasses the cutter blade **30**, and the other side of the diameter-enlarged hole portion forming device **10** is defined as the proximal end side. However, the arrangement of the diameter-enlarged hole portion forming device **10** is not limited to this.

(17) <Configuration of Components>

(18) As shown in FIG. 3A and FIG. 3B, the shank **20** includes an attachment portion **21** and an exterior portion **22**. The attachment portion **21** is in the shape of, for example, a rectangular column or a circular column. The attachment portion **21** is provided such that it is closer to the proximal end than the exterior portion **22** is. The attachment portion **21** and the exterior portion **22** are integrated together such that the attachment portion **21** and the exterior portion **22** are concentric with each other.

(19) The exterior portion **22** has a columnar shape, and includes a first insertion hole **23**, a communication hole **24**, and elongated holes **25**. The first insertion hole **23** has a columnar shape, and extends in the axial direction inside the exterior portion **22**. The first insertion hole **23** is open at the distal end side of the exterior portion **22**, and is bottomed at the proximal end side of the exterior portion **22**.

(20) For example, the communication hole **24** extends from the first insertion hole **23** in a manner to penetrate through one side of the exterior portion **22** in a direction orthogonal to the axial direction, and the communication hole **24** is provided such that it is closer to the distal end than the elongated holes **25** are. For example, the elongated holes **25** extend from the first insertion hole **23** in a manner to penetrate through both sides of the exterior portion **22**, respectively, in a direction orthogonal to the axial direction, and each of the elongated holes **25** is elongated in the axial

direction.

(21) The outer peripheral surface of the exterior portion **22** is provided with a first cylindrical groove **26**. The first cylindrical groove **26** is a groove to which an O-ring **70** (FIG. 1A) is attached. The first cylindrical groove **26** is disposed between the communication hole **24** and the elongated holes **25** in the axial direction. The first cylindrical groove **26** extends in an annular manner about the center axis of the exterior portion **22**. The first cylindrical groove **26** is recessed from the outer peripheral surface of the exterior portion **22**, and is open at the outer peripheral surface of the exterior portion **22**.

(22) The distal end of the exterior portion **22** is provided with a locking portion. For example, the locking portion is constituted by a pair of notches **27**. The pair of notches **27** is disposed such that the first insertion hole **23** is interposed between the notches **27** in a direction orthogonal to the axial direction. Each notch **27** is recessed from the distal end of the exterior portion **22** toward the proximal end side.

(23) As shown in FIG. 4A and FIG. 4B, the cutter blade **30** includes a coupling portion **31** and a pair of cutting portions **32**. It should be noted that the cutter blade **30** may include only one cutting portion **32** or three or more cutting portions **32**.

(24) The coupling portion **31** has a cylindrical shape. The coupling portion **31** includes a second insertion hole **33**, which penetrates through the inside of the coupling portion **31** in the axial direction. The proximal end of the coupling portion **31** is provided with a locked portion. For example, the locked portion is constituted by a pair of protrusions **34**. The protrusions **34** protrude from the proximal end of the coupling portion **31**.

(25) The pair of cutting portions **32** is disposed such that the cutting portions **32** are arranged parallel to each other with the axis of the coupling portion **31** interposed therebetween, and such that the inner surfaces of the respective cutting portions **32** face each other. The cutting portions **32** are plated-shaped, and protrude from the distal end of the coupling portion **31** in the axial direction.

(26) The cutting portions **32** are curved in the circumferential direction about the axis of the coupling portion **31**, and the inner surfaces of the respective cutting portions **32** are curved surfaces. The cutting portions **32** are flexible in a direction orthogonal to the inner surfaces, and the pair of cutting portions **32** can be deflected such that the distance therebetween is changeable.

(27) The distal ends of the cutting portions **32** are provided with respective cutting edges **35**. The cutting edges **35** are cutting tips, and the cutting edge **35** of each cutting portion **32** is disposed at the center of the cutting portion **32** in the circumferential direction. The cutting edge **35** protrudes from each of the following surfaces of the cutting portion **32**: the inner surface; the other side surface, i.e., the outer surface; and the distal end surface.

(28) As shown FIG. 5A to FIG. 5C, the bow jaw **40** includes a shaft **41** and a guide **42**, and the bow jaw **40** is provided with a suction passage **80** formed therein. The shaft **41** is a columnar rod-shaped component, and the guide **42** has a substantially columnar shape. The shaft **41** is disposed such that it is closer to the proximal end than the guide **42** is. The shaft **41** and the guide **42** are integrated together such that shaft **41** and the guide **42** are concentric with each other.

(29) The shaft **41** includes a first outer peripheral surface and a first pin hole **43**. The first outer peripheral surface surrounds the center axis between the distal end and the proximal end of the shaft **41**. The first pin hole **43** is disposed such that it is closer to the proximal end than the suction passage **80** is. The first pin hole **43** penetrates through the shaft **41** in a direction orthogonal to the axial direction.

(30) The proximal end of the guide **42** is connected to the shaft **41**, and the distal end of the guide **42** includes a distal end surface **44**. The guide **42** includes a second outer peripheral surface that surrounds the center axis between the distal end and the proximal end of the guide **42**. The second outer peripheral surface is provided with a pair of sloped surfaces **45** and a pair of recesses **46**.

(31) Although the bow jaw **40** is provided with the pair of sloped surfaces **45** in this example, the bow jaw **40** may alternatively be provided with only one sloped surface **45** or three or more sloped

surfaces **45**. In such a case, the cutter blade **30** is provided with the same number of cutting portions **32** as the number of sloped surfaces **45**. Although the bow jaw **40** is provided with the pair of recesses **46** in this example, the bow jaw **40** may be provided with only one recess **46** or three or more recesses **46**.

(32) The sloped surfaces **45** are surfaces on which the respective cutting portions **32** of the cutter blade **30** slide. For example, the sloped surfaces **45** are flat surfaces. As shown in FIG. **4B**, at a position where each cutting portion **32** faces the corresponding sloped surface **45**, the inner surface of the cutting portion **32** is curved in a manner to protrude away from the sloped surface **45**. Consequently, a gap **48** is formed between each cutting portion **32** and the corresponding sloped surface **45**.

(33) As shown in FIG. **5A** to FIG. **5C**, the sloped surfaces **45** are arranged between a pair of protrusions **47** in a direction orthogonal to the axial direction, and extend in the axial direction. The pair of protrusions **47** protrudes from the sloped surfaces **45**, and extends in the axial direction along the sloped surfaces **45**.

(34) Each sloped surface **45** is sloped from the shaft **41** side toward the distal end surface **44** side in a direction away from the center axis of the shaft **41**. The diameter of the guide **42** is increased from the proximal end side toward the distal end side by the sloped surfaces **45**. The pair of sloped surfaces **45** is disposed such that the axis is interposed therebetween. The distance between the pair of sloped surfaces **45** increases from the proximal end side toward the distal end side.

(35) Each recess **46** is disposed between the pair of sloped surfaces **45** in the circumferential direction, and is open at both the second outer peripheral surface and the distal end surface **44** of the guide **42**. Accordingly, each recess **46** is recessed to form a first opening **46a** in the second outer peripheral surface in a direction orthogonal to the axial direction, and is also recessed to form a second opening **46b** in the distal end surface **44** in the axial direction. The pair of recesses **46** is disposed such that the axis is interposed therebetween, and the suction passage **80** is disposed between the pair of recesses **46**.

(36) The suction passage **80** includes a flow passage **81**, a pair of first branch passages **82**, and a pair of second branch passages **83**. Alternatively, the passage **80** may include only one first branch passage **82**, and may include only one second branch passage **83**. Further alternatively, the suction passage **80** may include three or more first branch passages **82**, and may include three or more second branch passages **83**, in accordance with the number of sloped surfaces **45** and the number of recesses **46**.

(37) For example, the flow passage **81** branches into the first branch passages **82** and the second branch passages **83**. Accordingly, the suction passage **80** includes the following suction ports: a first suction port **81a** of the flow passage **81**; second suction ports **82a** of the first branch passages **82**; and third suction ports **83a** of the second branch passages **83**.

(38) For example, the diameter of the flow passage **81**, the diameter of each first branch passage **82**, and the diameter of each second branch passage **83** are the same. The suction passage **80** includes a discharge port **84** whose area is set to be greater than or equal to the sum of the area of the first suction port **81a**, the area of the second suction ports **82a**, and the area of the second suction ports **82a**.

(39) The flow passage **81** extends from the first suction port **81a** to the shaft **41** in the axial direction in a manner to penetrate through the guide **42**. The flow passage **81** is provided concentrically with the guide **42** and the shaft **41**. The first suction port **81a** of the flow passage **81** is open at the center of the distal end surface **44** of the guide **42**. The discharge port **84** of the flow passage **81** is open at the first outer peripheral surface of the shaft **41**, and the discharge port **84** is elongated to be longer in the axial direction than in the radial direction.

(40) The first branch passages **82** allow the flow passage **81** and the recesses **46** of the guide **42** to communicate with each other. For example, the first branch passages **82** are branched off from the flow passage **81**, and extend in a direction crossing (e.g., orthogonal to) the axial direction. Each of

the first branch passages **82** is open at the bottom of a corresponding one of the recesses **46**. For example, the second suction port **82a** of each first branch passage **82** may be disposed such that the position of the second suction port **82a** is shifted from the center of the corresponding recess **46** in the axial direction toward the distal end side.

(41) The pair of first branch passages **82** is arranged such that the first branch passages **82** are connected to each other via the flow passage **81** and extend in a straight line. In this case, by drilling a straight hole connecting between the pair of recesses **46**, the pair of first branch passages **82** can be formed at the same time, and thus the pair of first branch passages **82** can be readily formed.

(42) At the positions where the cutting portions **32** face the sloped surfaces **45**, the second branch passages **83** allow the gaps **48**, which are formed between the cutting portions **32** and the sloped surfaces **45** (see FIG. **4B**), and the flow passage **81** to communicate with each other. For example, the second branch passages **83** are branched off from the flow passage **81**, extend in a direction crossing (e.g., orthogonal to) the axial direction, and are open at the sloped surfaces **45**.

(43) The pair of second branch passages **83** is arranged such that the second branch passages **83** are connected to each other via the flow passage **81** and extend in a straight line. In this case, by drilling a straight hole connecting between the pair of sloped surfaces **45**, the pair of second branch passages **83** can be formed at the same time, and thus the pair of second branch passages **83** can be readily formed.

(44) The first branch passages **82** and the second branch passages **83** are provided at the same position in the axial direction. For this reason, these branch passages are arranged in a manner to cross each other (e.g., orthogonally). Alternatively, the first branch passages **82** and the second branch passages **83** may be arranged such that the positions of the first branch passages **82** and the positions of the second branch passages **83** are shifted from each other in the axial direction. Further alternatively, the pair of first branch passages **82** may be arranged such that the positions of the first branch passages **82** are shifted from each other, and the pair of second branch passages **83** may be arranged such that the positions of the second branch passages **83** are shifted from each other.

(45) The distal end surface **44** is expanded in a direction orthogonal to the axial direction, and the center portion of the distal end surface **44** protrudes. The center portion of the distal end surface **44** forms a flat plane, and the distal end surface **44** is sloped from the center portion to its outer periphery toward the proximal end side.

(46) The distal end surface **44** is provided with the first suction port **81a** of the flow passage **81**, the second openings **46b** of the pair of recesses **46**, a pair of first grooves **90**, and a pair of second grooves **91**. Alternatively, the distal end surface **44** may be provided with only one first groove **90**. Also, the distal end surface **44** may be provided with only one second groove **91**.

(47) The first suction port **81a** is disposed at the center of the distal end surface **44**. The pair of second openings **46b** is disposed such that the first suction port **81a** is interposed therebetween. Accordingly, the distal end surface **44** is provided with the first suction port **81a** and the pair of second openings **46b**, which are arranged in a straight line.

(48) The first grooves **90** are recessed from the distal end surface **44**, extend in a direction orthogonal to the axial direction, and are connected to the respective second openings **46b** of the recesses **46** and the first suction port **81a**. The pair of first grooves **90** is disposed such that one ends the respective first grooves **90** are connected to each other, and the other ends of the respective first grooves **90** are connected to the pair of second openings **46b**, respectively. The first grooves **90** form a straight line.

(49) The second grooves **91** are recessed from the distal end surface **44**, and extend in a manner to cross (e.g., orthogonally to) the first grooves **90** such that the second grooves **91** are connected to the first suction port **81a**. For example, the second grooves **91** extend from the first suction port **81a** to the respective sloped surfaces **45**, and are open at the respective sloped surfaces **45** as

groove openings **91a**.

(50) The depth of each second groove **91** from the distal end surface **44** is set such that, in a state where the cutting portions **32** have slid on the respective sloped surfaces **45** toward the distal end side to the greatest degree, the bottoms of the respective second grooves **91** are positioned closer to the distal end than the cutting portions **32** are. Accordingly, even when the cutting portions **32** have slid on the sloped surfaces **45**, the cutting portions **32** are positioned closer to the proximal end than the second grooves **91** are. Therefore, the groove openings **91a** of the second grooves **91** are not blocked by the cutting portions **32**.

(51) When the cutting portions **32** slide on and face the respective sloped surfaces **45**, the gaps **48** formed between the cutting portions **32** and the sloped surfaces **45** are positioned near the groove openings **91a** of the second grooves **91**.

(52) As shown in FIG. 1A, FIG. 2A, and FIG. 6A, the stopper sleeve **50** has a cylindrical shape, and includes a mounting hole **51**, a second pin hole **52**, and a first fixing hole **53**. The mounting hole **51** penetrates through the stopper sleeve **50** in the axial direction of the stopper sleeve **50**. The diameter of the mounting hole **51** is set to be greater than the external diameter of the shank **20** so that the shank **20** can be inserted into the mounting hole **51**.

(53) The second pin hole **52** is a hole in which a pin **71** is inserted. The second pin hole **52** extends from the mounting hole **51** in a direction orthogonal to the axial direction, penetrating through the stopper sleeve **50**. The first fixing hole **53** is a hole in which a screw for fixing the pin **71** to the stopper sleeve **50** is inserted. The first fixing hole **53** extends in a direction orthogonal to the axial direction, and crosses (e.g., orthogonally to) the second pin hole **52**.

(54) As shown in FIG. 1A and FIG. 2A, the suction adapter **60** includes an annular part **61** and a pipe part **62**. The annular part **61** has a cylindrical shape, and includes a shouldered hole **63** and a second cylindrical groove **64**. The shouldered hole **63** penetrates through the annular part **61** in the axial direction thereof, and includes a large-diameter portion **65** having a large diameter. The diameter of the large-diameter portion **65** is greater than the external diameter of the shank **20**, and except the large-diameter portion **65**, the diameter of the shouldered hole **63** is substantially equal to the external diameter of the shank **20**.

(55) The second cylindrical groove **64** is a groove to which the O-ring **70** is attached. The second cylindrical groove **64** is open at the inner surface of the annular part **61**, and is disposed such that the second cylindrical groove **64** is closer to the proximal end than the large-diameter portion **65** is. The pipe part **62** is coupled to the annular part **61** such that the pipe part **62** is in communication with the shouldered hole **63** of the annular part **61**.

(56) <Assembling of Diameter-Enlarged Hole Portion Forming Device>

(57) As shown in FIG. 1A and FIG. 2A, the protrusions **34** of the cutter blade **30** are fitted to the notches **27** of the shank **20**. As a result, the protrusions **34** are locked to the notches **27**, and the proximal end of the cutter blade **30** and the distal end of the shank **20** are connected to each other. The cutter blade **30** and the shank **20** are arranged concentrically with each other, and the second insertion hole **33** of the cutter blade **30** and the first insertion hole **23** of the shank **20** communicate with each other.

(58) Next, a spring **73** is inserted in the first insertion hole **23**, and then the shaft **41** of the bow jaw **40** is inserted in the first insertion hole **23** and the second insertion hole **33**. As a result, the spring **73** is disposed between the proximal end of the shaft **41** and the proximal end of the first insertion hole **23**.

(59) Subsequently, the shank **20** is inserted in the mounting hole **51** of the stopper sleeve **50**. At the time, the stopper sleeve **50**, the shank **20**, and the bow jaw **40** are arranged such that the second pin hole **52** of the stopper sleeve **50**, the elongated holes **25** of the shank **20**, and the first pin hole **43** of the bow jaw **40** coincide with each other in a direction orthogonal to the axial direction.

(60) Then, the pin **71** is inserted in the second pin hole **52**, the elongated holes **25**, and the first pin hole **43**. Thereafter, a screw is inserted in, and screwed with, a second fixing hole **72** of the pin **71**

and the first fixing hole **53** of the stopper sleeve **50**. As a result, the pin **71** is fixed to the stopper sleeve **50**.

(61) In the axial direction, the dimension of the pin **71** is substantially the same as the dimension of each of the first pin hole **43** and the second pin hole **52** while the elongated holes **25** are longer in dimension than the pin **71**, the first pin hole **43**, and the second pin hole **52**. Accordingly, the pin **71** is fixed to the first pin hole **43** and the second pin hole **52**, but movable in the axial direction relative to the elongated hole **25**.

(62) Consequently, the shank **20** and the cutter blade **30**, and the bow jaw **40**, are movable relative to each other in the axial direction, and thereby the diameter-enlarged hole portion forming device **10** is extendable/contractible. When the diameter-enlarged hole portion forming device **10** contracts, the cutter blade **30** moves toward the distal end side relative to the bow jaw **40**. At the time, the more the cutter blade **30** moves toward the distal end side, the more the spring **73** urges the cutter blade **30** toward the proximal end side.

(63) Further, the shank **20** is inserted in the shouldered hole **63** of the suction adapter **60** to mount the suction adapter **60** to the shank **20**. As a result, the inner surface of the annular part **61** and the outer surface of the shank **20** face each other, and the large-diameter portion **65** forms a connecting passage **66** between these surfaces. The connecting passage **66** is formed in an annular manner so as to surround the shank **20**.

(64) At the time, the large-diameter portion **65** of the shouldered hole **63**, the communication hole **24** of the shank **20**, and the discharge port **84** of the suction passage **80** are arranged such that they coincide with each other in a direction orthogonal to the axial direction. As a result, the suction passage **80** communicates with the annular connecting passage **66** via the communication hole **24**. The O-ring **70** fitted to the first cylindrical groove **26** and the second cylindrical groove **64** retains the shank **20** such that the shank **20** is rotatable relative to the suction adapter **60**.

(65) <Operation of Diameter-Enlarged Hole Portion Forming Device>

(66) The diameter-enlarged hole portion forming device **10** forms a diameter-enlarged hole portion E in a pre-formed hole H of a workpiece B. The pre-formed hole H is formed in the workpiece B, which is, for example, a concrete wall, by a hole forming machine, such as a drill. For example, the pre-formed hole H is a bottomed hole having a columnar shape.

(67) A rotation motor or the like is connected to the attachment portion **21** of the shank **20**, and the rotation motor or the like is started. As a result, the shank **20**, and the cutter blade **30** and the bow jaw **40** coupled to the shank **20**, rotate. Also, a suction machine is connected to the pipe part **62** of the suction adapter **60**, and the suction machine is started. At the time, since the shank **20** and the suction adapter **60** are not fixed to each other, the shank **20** rotates, but the suction adapter **60** does not rotate.

(68) The diameter-enlarged hole portion forming device **10** is inserted into the pre-formed hole H. At the time, since the cutter blade **30** is in the state of being urged by the spring **73** toward the proximal end side of the bow jaw **40**, the cutting portions **32** of the cutter blade **30** are positioned at the proximal end side of the sloped surfaces **45** of the bow jaw **40**. Accordingly, the distance between the pair of cutting portions **32** is less than the diameter of the opening of the pre-formed hole H, and the cutter blade **30** and the bow jaw **40** can be inserted into the pre-formed hole H.

(69) Then, the cutter blade **30** is pushed to the bow jaw **40** toward the distal end side. As a result, the cutting portions **32** of the cutter blade **30** move on the sloped surfaces **45** of the bow jaw **40** toward the distal end side. It should be noted that, at the time, the cutting portions **32** slide between and along the pair of protrusions **47**, and are guided by the pair of protrusions **47** in the axial direction.

(70) Then, the cutting portions **32** are deflected along the sloped surfaces **45**, and the distance between the pair of cutting portions **32** increases. At the same time, the cutting portions **32** rotate. Consequently, the cutting edges **35** of the cutting portions **32** come into contact with, and cut into, the inner surface of the pre-formed hole H, and thereby the diameter-enlarged hole portion E is

formed in the pre-formed hole H. At the time, powder dust is generated due to the cutting of the workpiece.

(71) The powder dust enters the recesses **46**, which are adjacent to the cutting edges **35** in the rotation direction, and is sucked into the first branch passages **82** through the second suction ports **82a**, which are open in the recesses **46**. Here, the powder dust that is not sucked in through the second suction ports **82a** and remains in the recesses **46** moves downward inside the recesses **46** due to its own weight, reaches the second openings **46b**, passes through the first grooves **90** connected to the second openings **46b** to be led to the first suction port **81a**, and is sucked into the flow passage **81** through the first suction port **81a**.

(72) The powder dust also moves downward from the cutting edges **35**, or enters the gaps **48** between the cutting portions **32** and the sloped surfaces **45**. In these cases, the powder dust moves downward, and is sucked into the second branch passages **83** through the third suction ports **83a**, which are positioned below the cutting edges **35** and the gaps **48**. Here, the powder dust that is not sucked in through the third suction ports **83a** moves downward along the sloped surfaces **45**, reaches the groove openings **91a**, passes through the second grooves **91** from the groove openings **91a** to be led to the first suction port **81a**, and is sucked into the flow passage **81** through the first suction port **81a**.

(73) As described above, in the guide **42**, the second suction ports **82a** and the third suction ports **83a** are arranged such that they are closer to the cutting edges **35** than the first suction port **81a** is. Accordingly, the powder dust that is generated due to the cutting by the cutting edges **35** can be sucked in more speedily. Moreover, since the first suction port **81a** is disposed below the second suction ports **82a** and the third suction ports **83a**, the powder dust that moves downward without being sucked in through the second suction ports **82a** and the third suction ports **83a** can be sucked in through the first suction port **81a**. Furthermore, the first grooves **90** lead the powder dust of the recesses **46** to the first suction port **81a**, and the second grooves **91** lead the powder dust of the sloped surfaces **45** and the gaps **48** to the first suction port **81a**. Consequently, the powder dust can be sucked in more efficiently.

(74) The powder dust that is sucked in through each suction port in the above-described manner passes through the suction passage **80**, then passes through the communication hole **24** and the connecting passage **66** from the discharge port **84** of the suction passage **80**, and is then collected into the suction machine through the pipe part **62**.

(75) As described above, the powder dust that is generated during the cutting work is sucked in through a plurality of suction ports, and also, the powder dust is led by the recesses **46** and the grooves to be sucked in through the first suction port **81a**. This makes it possible to reduce locking, in which the rotation of the diameter-enlarged hole portion forming device **10** stops due to the powder dust getting caught between components of the device **10**, and thereby the cutting work can be performed smoothly.

(76) In addition, the diameter-enlarged hole portion forming device **10** need not be provided with a mechanism for releasing the locking. This makes it possible to simplify, and reduce the cost of, the diameter-enlarged hole portion forming device **10**. Since the powder dust that is scattered to the outside during the cutting work can be reduced, the worker's exposure to the powder dust is reduced, which makes it possible to improve the work environment.

(77) Further, since the first suction port **81a** faces the bottom of the pre-formed hole H, the powder dust that has fallen on the bottom of the pre-formed hole H can be sucked in through the first suction port **81a**. For this reason, when the cutting work is done, only a small amount of powder dust remains inside the pre-formed hole H and the diameter-enlarged hole portion E. This makes the cleaning of the remaining powder dust easy, or even makes it possible to eliminate the necessity of the cleaning of the remaining powder dust.

(78) As a result of the diameter-enlarged hole portion E being formed in the pre-formed hole H, for example, a bolt hole is formed in the workpiece B. An anchor bolt is inserted into the bolt hole, and

then the head of the anchor bolt is expanded. Consequently, the expanded head of the anchor bolt is locked to the diameter-enlarged hole portion E, and thereby the pull-out strength of the anchor bolt increases.

(79) <Variation 1>

(80) As shown in FIG. 7A to FIG. 7C, the diameter-enlarged hole portion forming device **10** according to Variation 1 includes a bow jaw **140**, in which the guide **42** need not include the second grooves **91**. Other than this point, the diameter-enlarged hole portion forming device **10** according to Variation 1 is the same in configuration as the diameter-enlarged hole portion forming device **10** of FIG. 1A. Accordingly, the guide **42** includes the recesses **46** and the first grooves **90**, and the suction passage **80** includes the flow passage **81**, the first branch passages **82**, and the second branch passages **83**. Therefore, the diameter-enlarged hole portion forming device **10** can efficiently remove the powder dust.

(81) It should be noted that in the diameter-enlarged hole portion forming device **10** according to Variation 1, at least either the first branch passages **82** or the second branch passages **83** may be eliminated from the suction passage **80**. Even in such a case, the powder dust can be removed efficiently by the flow passage **81**.

(82) <Variation 2>

(83) As shown in FIG. 8A to FIG. 8C, the diameter-enlarged hole portion forming device **10** according to Variation 2 includes a bow jaw **240**, in which the guide **42** need not include the first grooves **90** and the second grooves **91**. Other than this point, the diameter-enlarged hole portion forming device **10** according to Variation 2 is the same in configuration as the diameter-enlarged hole portion forming device **10** of FIG. 1A. Accordingly, the guide **42** includes the recesses **46**, and the suction passage **80** includes the flow passage **81**, the first branch passages **82**, and the second branch passages **83**. Therefore, the diameter-enlarged hole portion forming device **10** can efficiently remove the powder dust.

(84) <Variation 3>

(85) As shown in FIG. 9A to FIG. 9C, the diameter-enlarged hole portion forming device **10** according to Variation 3 includes a bow jaw **340**, in which the guide **42** need not include the first grooves **90** and the second grooves **91**, and the suction passage **80** need not include the second branch passages **83**. Other than this point, the diameter-enlarged hole portion forming device **10** according to Variation 3 is the same in configuration as the diameter-enlarged hole portion forming device **10** of FIG. 1A. Accordingly, the guide **42** includes the recesses **46**, and the suction passage **80** includes the flow passage **81** and the first branch passages **82**. Therefore, the diameter-enlarged hole portion forming device **10** can efficiently remove the powder dust.

(86) It should be noted that in the diameter-enlarged hole portion forming device **10** according to Variation 3, the suction passage **80** may include the second branch passages **83** without including the first branch passages **82**. Alternatively, in the diameter-enlarged hole portion forming device **10** according to Variation 3, the suction passage **80** need not include the first branch passages **82** and the second branch passages **83**. Even in each of these cases, the powder dust can be removed efficiently by the flow passage **81**.

(87) <Other Variations>

(88) The guide **42** of the bow jaw **140** of the diameter-enlarged hole portion forming device **10** according to another variation may include the second grooves **91** without including the first grooves **90**. In such a case, the suction passage **80** may include both the first branch passages **82** and the second branch passages **83**, or alternatively, at least either the first branch passages **82** or the second branch passages **83** may be eliminated from the suction passage **80**. Even in each of these cases, the powder dust can be removed efficiently by the flow passage **81**.

(89) In the bow jaw **140** of the diameter-enlarged hole portion forming device **10** according to yet another variation, the guide **42** may include the first grooves **90** and the second grooves **91**, and at least either the first branch passages **82** or the second branch passages **83** may be eliminated from

the suction passage **80**. Even in such a case, the powder dust can be removed efficiently by the flow passage **81**.

(90) In all of the above-described diameter-enlarged hole portion forming devices **10**, the guide **42** need not include the recesses **46**. Even in such a case, the powder dust can be removed efficiently by the flow passage **81**.

(91) In all of the above-described diameter-enlarged hole portion forming devices **10**, corners between the distal end surface **44** and the first grooves **90**, and corners between the distal end surface **44** and the second grooves **91**, may be chamfered.

(92) In all of the above-described diameter-enlarged hole portion forming devices **10**, the proportion of the suction force at the first suction port **81a**, the suction force at the second suction ports **82a**, and the suction force at the third suction ports **83a** can be controlled by adjusting the diameters of the flow passage **81**, the first branch passages **82**, and the second branch passages **83** and the cross-sectional areas of the first grooves **90** and the second grooves **91**.

INDUSTRIAL APPLICABILITY

(93) The diameter-enlarged hole portion forming device according to the present invention is useful as, for example, a diameter-enlarged hole portion forming device capable of efficiently removing powder dust.

REFERENCE CHARACTERS LIST

(94) **10**: diameter-enlarged hole portion forming device **30**: cutter blade **32**: cutting portion **35**: cutting edge **40**: bow jaw **41**: shaft **42**: guide **44**: distal end surface **45**: sloped surface **46**: recess **80**: suction passage **81**: flow passage **81a**: first suction port (suction port) **82**: first branch passage **83**: second branch passage **90**: first groove **91**: second groove **140**: bow jaw **240**: bow jaw **340**: bow jaw

Claims

1. A diameter-enlarged hole portion forming device for forming a diameter-enlarged hole portion by cutting into a hole that is formed in a workpiece, the diameter-enlarged hole portion forming device comprising: a bow jaw including a shaft and a guide, the guide including a distal end surface and a first sloped surface, the distal end surface being provided on an opposite side of the guide from the shaft, the first sloped surface being sloped from the shaft side of the guide toward the distal end surface side of the guide in a direction away from a center axis of the shaft, wherein the guide includes: a pair of sloped surfaces that includes the first sloped surface and a second sloped surface, wherein the pair of the sloped surfaces is provided such that the center axis is interposed between the pair of the sloped surfaces; a recess that is open at the distal end surface and an outer peripheral surface between the pair of the sloped surfaces; and a groove that is connected to a suction port and to an opening of the recess, wherein the opening is formed at the distal end surface, and the suction port is open at the distal end surface; and a cutter blade including a cutting portion whose distal end is provided with a cutting edge, the cutting portion being slidable along the first sloped surface, wherein the bow jaw further includes a suction passage that includes a flow passage, the flow passage including the suction port.
2. The diameter-enlarged hole portion forming device according to claim 1, wherein the suction passage includes a branch passage that is branched off from the flow passage and that communicates with the recess.
3. The diameter-enlarged hole portion forming device according to claim 1, wherein the suction passage includes a branch passage that is branched off from the flow passage and that is open at the first sloped surface.
4. A diameter-enlarged hole portion forming device for forming a diameter-enlarged hole portion by cutting into a hole that is formed in a workpiece, the diameter-enlarged hole portion forming device comprising: a bow jaw including a shaft and a guide, the guide including a distal end surface

and a sloped surface, the distal end surface being provided on an opposite side of the guide from the shaft, the sloped surface being sloped from the shaft side of the guide toward the distal end surface side of the guide in a direction away from a center axis of the shaft, wherein the guide includes a groove that is open at the sloped surface and that is connected to a suction port in the distal end surface; and a cutter blade including a cutting portion whose distal end is provided with a cutting edge, the cutting portion being slidable along the sloped surface, wherein the bow jaw further includes a suction passage that includes a flow passage, the flow passage including the suction port.

5. The diameter-enlarged hole portion forming device according to claim 4, wherein the suction passage includes a branch passage that is branched off from the flow passage and that is open at the sloped surface.
